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Cervical cancer burden in India: A descriptive epidemiological study and policy insights

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ABSTRACT

Background: Cervical cancer remains a major global health issue, particularly in low- and middle-income countries (LMICs). Although human papillomavirus (HPV) vaccination and screening are proven preventive strategies, LMICs, including India, face significant implementation challenges.

Methods: This observational, descriptive epidemiological study analyzes cervical cancer burden across WHO regions sourcing data from GLOBOCAN 2022, GBD, and GHO. India-specific state-level analysis was conducted using GBD data. Joinpoint regression assessed death trends, and a comparative analysis examined the impact of India's 2016 national cervical cancer screening and management policies.

Findings: The Southeast Asia Region (SEARO) accounts for the second-highest cervical cancer incident (new cases) and death rate among WHO regions, with India contributing over 65 % of the burden. National screening coverage remains alarmingly low, with only 1.9 % of women aged 30–49 undergoing screening, far below developed nations. Despite the adoption of Visual Inspection with Acetic Acid (VIA) as primary screening method in 2016, India's cervical cancer death rates have continued to rise, increasing from 6.06 to 6.78 per 100,000 women (2012–2016) to 6.82–6.91 (2016–2019). However, death annual percentage change declined from 3.84 % (2012–2015) to 0.46 % (2016–2019), indicates slowdown in death acceleration but not a reversal.

Conclusion: India's burden remains high due to low screening coverage, reliance on subjective screening test, and limited HPV vaccination. While many countries like Australia and Bhutan have successfully reduced incidence and death through HPV-based screening and vaccination, India's slow progress underscores the urgent need for policy shifts towards HPV-DNA testing with self-sampling option and national HPV-vaccination programs implementation to curb cervical cancer burden effectively.

Introduction

Cervical cancer, a preventable disease, is a significant global public health problem that disproportionately affects women in low- and middle-income countries (LMICs). Global Cancer Observatory (GLOBOCAN 2022) report estimated 662,301 new cervical cancer cases and

Abbreviations: AB-PMJAY, Ayushman Bharat—Pradhan Mantri Jan Aarogya Yojana; AFRO, African Region; APC, Annual Percentage Change; ASHA, Accredited Social Health Activist; ASR, Age-Standardized Ratio; CDC, Centers for Disease Control and Prevention; CIR, Crude Incidence Rate; CMR, Crude Mortality Rate; COVID, Coronavirus Disease; DALY, Disability-Adjusted Life Year; HTA1n, Health Technology Assessment India (HTA1n); DNA, Deoxyribonucleic Acid; EMRO, Eastern Mediterranean Region; EURO, European Region; GAVI, Global Alliance for Vaccines and Immunization; GBD, Global Burden of Disease; GHO, Global Health Observatory; GLOBOCAN, Global Cancer Observatory; HC2, Hybrid Capture 2; HFW, Health and Family Welfare; HPV, Human Papillomavirus; ICMR, Indian Council of Medical Research; IEC, Information, Education, and Communication; Incidence, New cases; INR, Indian rupee (Currency); LMIC, Low- and Middle-Income Country; MoHFW, Ministry of Health and Family Welfare; Mortality, Death; NCD, Non-Communicable Disease; NFHS, National Family Health Survey; NGO, Non Government Organization; NHFS, National Health Facility Survey; NHM, National Health Mission; NIP, National Immunization Program; NP-NCDC, National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke—National Centre for Disease Control; PAHO, Pan American Health Organization; PAP Smear, Papanicolaou test; RT-PCR, Reverse Transcription Polymerase Chain Reaction; SEARO, South-East Asia Region; US, United States; USA, United States of America; VIA, Visual Inspection with Acetic Acid; WHO, World Health Organization; WPRO, Western Pacific Region; YLD, Years of Life Lost due to Disability.

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348,874 mortalities globally for 2022 [1,2]. Making cervical cancer the fifth most common cancer by new cases and the fourth by deaths worldwide. The situation in the South Asian region, particularly in India, is concerning; cervical cancer ranks as the second most common cancer among women in India, with an estimated 127,526 new cases and 79,906 deaths in 2022 [1,2]. The association between Human Papillomavirus (HPV) infection and cervical cancer is well established, with nearly 99 % of cervical cancer cases caused by persistent infection with high-risk HPV, particularly strains 16 and 18 [3]. Effective preventive measures include primary (HPV vaccination), secondary (screening, early detection & treatment), and tertiary (reducing disability) strategies. By integrating these measures into cervical cancer control policies, some developing countries have made progress in reducing the burden of cervical cancer [1,4]. On the contrary, most of the LMICs are struggling to reduce the morbidity and cervical cancer-related deaths [1]. Nearly 60–70 % of cervical cancer cases in LMICs, including India, are diagnosed at an advanced local stage, underscoring the urgent need for effective screening measures [5]. As many LMICs focus on secondary preventive strategies, the available screening tests include visual inspection with acetic acid (VIA), which has varied sensitivity and specificity as reported in by Osmanabad District Cervical Screening Study Group in 2005; Pap Smear (Papanicolaou test), which has good specificity but lower sensitivity (around 50–55 %); and HPV-DNA testing, which offers higher sensitivity (up to 98 %) but lower specificity [6–8]. Despite its high variability in sensitivity and specificity, attributed to multiple factors, including subjectivity, age-related differences among women, variability in interpretation skills, and the level of training and expertise of the examiner, VIA remains the most widely used screening test in India [9]. In addition to addressing the necessity of implementing HPV-DNA testing as a primary screening method and promoting HPV vaccination to lower the cervical cancer burden in India, this article aims to thoroughly evaluate the global burden of cervical cancer, with a particular focus on India and analyze the current cervical cancer screening and management practices.

Materials and methods

To position India's cervical cancer burden in a global context, we obtained regional Crude Incidence Rates (CIR) (new cases), Crude Mortality Rates (CMR) (deaths), Age-Standardized Rates (ASRs), and absolute case counts for all six World Health Organization (WHO) regions—from the Global Cancer Observatory (GLOBOCAN2022) [1]. Because India lacks a long-term, population-based state registry (and GLOBOCAN does not provide subnational mortality data), we extracted state-level cervical cancer mortality, Years Lived with Disability (YLD), and Disability-Adjusted Life Years (DALYs) for 2012–2019 from the Global Burden of Disease (GBD) online database [1,10]. By combining these sources, we aimed to achieve two specific goals: (1) compare India's 2022 cervical cancer burden against other WHO regions (and within SEARO), and (2) examine international trends over time. To project nationwide cervical cancer mortality patterns and annual percentage change (APC) in India, the Crude Mortality and female population data for the corresponding years 2012–2019 were obtained from the Global Health Observatory (GHO) [11]. While GLOBOCAN relies primarily on country-reported registries and extrapolation, GBD integrates surveys, verbal autopsies, and statistical models, leading to divergent mortality estimates for India; these methodological differences may limit direct comparability of mortality figures between the two databases [1,10]. Additionally, we searched the websites of the Press Information Bureau of India, Population Council, WHO, India's Ministry of Health and Family Welfare, PubMed, and official national health portals for relevant reports and guidelines. All data analyses were performed in Microsoft Excel 2021 and GraphPad Prism 8.0.2.

Statistical analysis

The global burden of cervical cancer across different regions with varying population structures were determined by ASRs sourced from the GLOBOCAN 2022. To evaluate the cervical cancer death burden in different states of India, all states were categorized into different regions (North, South, East, West, and Northeast), and the averages for death, YLD, and DALYs for two distinct time frames: 2012–2015 (pre-2016 guidelines) and 2016–2019 (post-2016 guidelines) were calculated for each region. To estimate temporal trends in cervical cancer mortality from 2012 to 2019, Joinpoint regression (version 4.9.1.0; US NCI Surveillance Research Program) was applied to state-level crude mortality rates (CMR per 100,000) using a piecewise log-linear model [2,12]. The estimated death trend, including APC for each Joinpoint, lower and upper confidence intervals, *P*-values, and Average Annual Percentage Change (AAPC) for cervical cancers, were presented [12]. As the Joinpoint regression (time trend) analysis relies on GBD data, changes in estimation methods and data quality over time may limit the comparability of statistics across different years. Further, to assess global guideline impact, we reviewed selected countries across WHO regions for the existence of national screening programs, HPV vaccination coverage and adoption of HPV testing as the primary screening modality.

Results

Cervical Cancer a looming threat to women's health globally

The WHO African Region (AFRO) faces the highest burden of cervical cancer among all regions with the highest Age-Standardized Rates (ASRs) of both new cases (31.8 per 100,000) and deaths (21.4 per 100,000) (Supplementary Table 1). In contrast, the WHO Eastern Mediterranean Region (EMRO) has the lowest burden, with the lower number of new cases (5.2) and death (3.4) ASRs per 100,000. The remaining WHO regions—Western Pacific (WPRO), Region of the Americas (PAHO), and Europe (EURO) all demonstrate lower new case and death ASRs compared to AFRO and SEARO. Among all WHO regions, the South-East Asia Region (SEARO) ranks second after the African region (AFRO) in terms of cervical cancer new cases (17.7) and death (10.8) ASRs per 100,000 people, respectively. Notably, the SEARO region contributes the maximum number of new cases (29.59 %) and deaths (34.12 %) across all WHO regions, as shown in Fig. 1A-B. Although women in AFRO have the highest individual risk (as reflected by ASRs), the total number of new cases (0.20 million) and deaths (0.12 million) are higher in SEARO, owing to its larger women population.

The SEARO region consists of eleven countries, all classified as upper or lower-middle-income except for South Korea (high-income) and Nepal (lower income). As evident from Table 1, the top five countries with the highest incidence or new case ASRs (23.3–14.9 per 100,000) are Indonesia, Maldives, Myanmar, India, and Thailand. Furthermore, these countries also have higher death ASRs (13.4–11.2 per 100,000 people). However, in the SEARO region, India contributes the highest number of new cervical cancer cases (127,526) and death (79,906), as shown in Fig. 1C-D, which highlights a significantly higher cervical cancer burden in India.

India's cervical cancer epidemic

As per the Global Burden of Disease data from 2012 to 2019 (Table 2), in India, the South Region bears the highest cervical cancer burden, with an average of 7.25 mortalities per 100,000 women from 2012 to 2015. In contrast, an increase in mortalities (7.6 per 100,000 women) was recorded from 2016 to 2019. Additionally, the West Region demonstrated the second-highest mortalities of 5.7 per 100,000 women during the years 2012–2015. Similarly, in the preceding four years (2016–2019), further high mortalities of 6.4 per 100,000 were recorded.

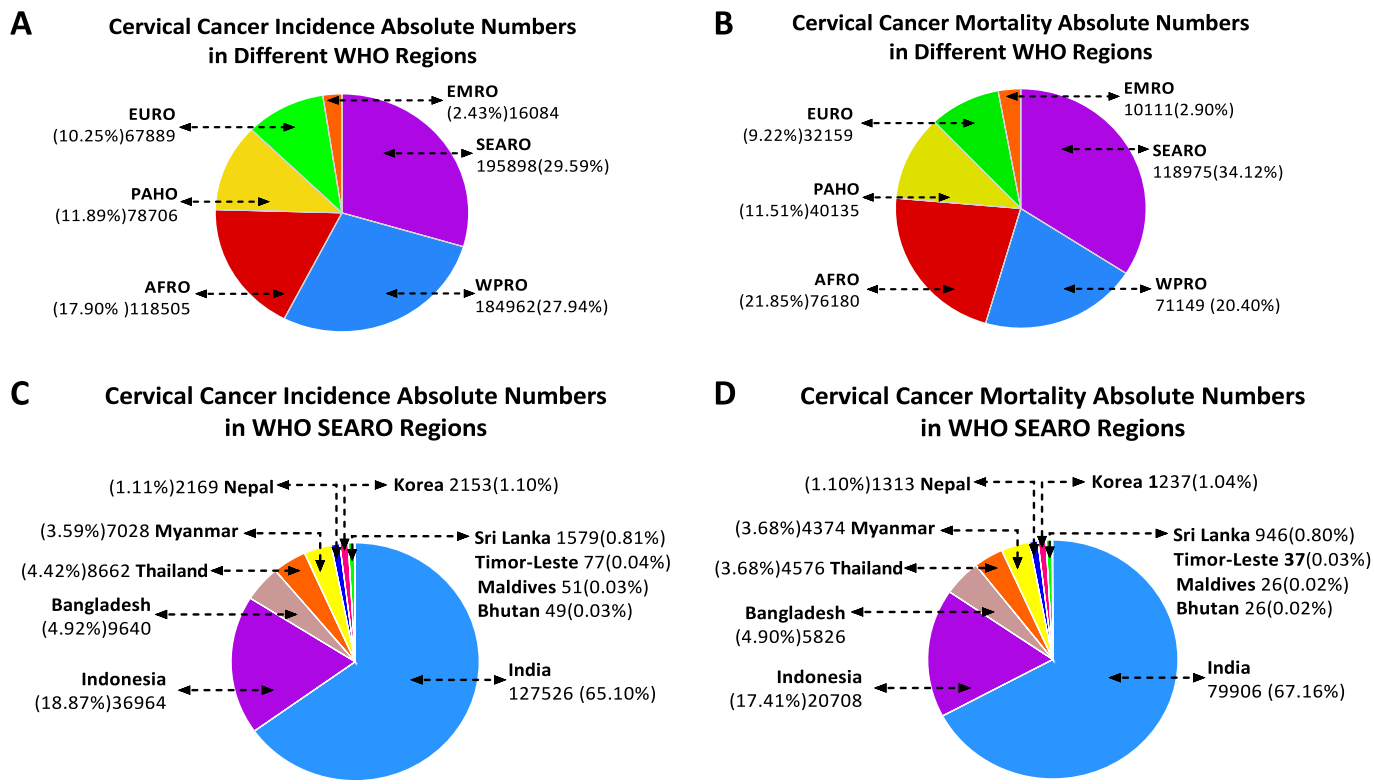

Fig. 1. Cervical cancer associated, A. Incidence (new cases) in absolute numbers across WHO regions, B. Mortality (deaths) in absolute numbers across WHO regions, C. Incidence (new cases) in absolute numbers within the WHO Southeast Asia Regional Office (SEARO), D. Mortality (deaths) in absolute numbers within the WHO SEARO *(Data Source: GLOBOCAN 2022 Cancer Today)*.

Table 1
Incidence and mortality Age-Standardized Ratios (ASRs) for cervical cancer in WHO SEARO region (Data Source: GLOBOCAN 2022).

	Incidence			Mortality				
Country Name	ASR (World)	Crude rate	Total Incidence	ASR (World)	Crude rate	Total Mortality	MIR in Percent	Percent Live
Indonesia	23.30	26.70	36,964	13.20	14.90	20,708	56.02	43.98
Maldives	22.60	25.10	51	12.90	12.80	26	50.98	49.02
Myanmar	21.40	24.60	7028	13.40	15.30	4374	62.24	37.76
India	17.70	18.90	127,526	11.20	11.80	79,906	62.66	37.34
Thailand	14.90	24.10	8662	6.90	12.70	4576	52.83	47.17
Timor-Leste	14.80	11.40	77	7.90	5.50	37	48.05	51.95
Nepal	14.20	13.40	2169	8.70	8.10	1313	60.53	39.47
Bhutan	13.60	13.30	49	7.20	7.10	26	53.06	46.94
Korea	11.80	16.20	2153	6.80	9.30	1237	57.45	42.55
Bangladesh	11.30	11.60	9640	7.00	7.00	5826	60.44	39.56
Sri Lanka	9.20	14.00	1579	5.30	8.40	946	59.91	40.09

Compared to the South region, the North Region demonstrated the second-lowest average mortalities, with 5.3 per 100,000 women from 2012 to 2015. Similar to other regions, a further increase in mortalities (5.6 per 100,000) was recorded from 2016 to 2019. The East Region of India exhibited an average of 5.65 mortalities per 100,000 women from 2012 to 2015. After introducing the cervical cancer screening & management policy in 2016 (operational guidelines), a slight decrease in average mortalities (5.6 per 100,000) was recorded in the following four years. Among all regions, the northeast region exhibited the lowest average mortalities (5.2) during the years 2012–2015. However, similar to other regions, a further increase in mortalities (5.4) was recorded from 2016 to 2019.

As evident from Table 2, the national average for Deaths, YLDs, and DALYs has increased from 5.8, 4.4, and 200.2 to 6.0, 4.6, and 205.5 per 100,000 women post-adaptation of cervical cancer screening using VIA test in NPCDCS (National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular diseases and Stroke) in 2016 in India [9]. In contrast, the multijoinpoint trend analysis (Supplementary Fig. 1 and Supplementary Table 2) reveals an increased cervical cancer CMR rate from 2012 to 2019 (6.0–6.8 between 2012 and 2016 to 6.8–6.9 between 2016 and 2019). While there was an increase in cervical cancer mortalities during 2012–2019, the annual percentage change (APC) in mortalities shifted from 3.844 % (2012–2015) to 0.5 % (2016–2019), representing a difference of 3.4 %. Furthermore, a similar pattern of increased cervical cancer-related mortalities in India can be verified from GLOBOCAN 2022 [1].

According to the fifth National Family Health Survey (NFHS5) conducted in India between 2019 and 2021, only 1.9 % of women between the ages of 30 and 49 have ever received a cervical cancer screening (2.2 % in urban regions and 1.7 % in rural areas) [13]. The studies conducted in India show that about 60–70 % of cervical cancer cases are diagnosed at an advanced stage with a poor prognosis [5]. As reported in research studies, reasons for lower cervical cancer screening rates in India, including limited access to infrastructure, especially in rural and remote

Table 2
Cervical cancer-related mortality rate, Years Lost to Disability (YLD), and Disability-Adjusted Life Years (DALYs) pre and post 2016 guidelines in India (Data Source: Global Burden of Disease).

Region	State	Year 2012–2015			year 2016–2019		
		Average Value			Average Value		
		Mortality	YLDs	DALYs	Mortality	YLDs	DALYs
North India	National Average	5.80	4.36	200.23	6.04	4.59	205.46
	Haryana	3.67	2.84	128.43	3.99	3.16	138.54
	Himachal Pradesh	6.96	5.11	233.07	7.02	5.20	229.59
	Jammu & Kashmir and Ladakh	2.58	2.08	88.95	2.73	2.27	92.70
	Punjab	6.66	4.98	224.39	7.51	5.71	249.73
	Uttar Pradesh	5.35	3.83	199.96	5.96	4.32	220.78
	Uttarakhand	6.48	4.87	232.77	6.31	4.81	220.94
	Average	**5.28**	**3.95**	**184.60**	**5.59**	**4.24**	**192.05**
South India	Andhra Pradesh	5.59	4.21	189.61	5.70	4.34	187.58
	Karnataka	10.05	7.07	340.40	11.21	8.02	378.53
	Kerala	6.01	4.77	183.83	6.41	5.07	192.54
	Tamil Nadu	12.26	8.32	399.99	12.54	8.80	403.28
	Telangana	4.76	4.14	169.98	4.70	4.22	163.95
	Other Union Territories	4.83	4.20	170.31	4.99	4.30	171.70
	Average	**7.25**	**5.45**	**242.35**	**7.59**	**5.79**	**249.60**
East India	Bihar	4.74	3.52	176.35	4.73	3.57	172.26
	Chhattisgarh	7.69	5.25	283.59	7.63	5.26	275.32
	Jharkhand	7.05	4.86	250.41	6.40	4.36	216.68
	Odisha	4.15	2.84	144.33	4.05	2.83	137.84
	West Bengal	4.64	3.69	162.73	5.01	4.12	173.28
	Average	**5.65**	**4.03**	**203.48**	**5.56**	**4.03**	**195.07**
West India	Goa	4.40	3.77	140.84	4.47	3.80	138.52
	Gujarat	5.74	4.34	202.17	6.71	5.26	235.08
	Madhya Pradesh	6.34	4.33	222.05	6.53	4.60	225.91
	Maharashtra	7.78	5.68	256.62	9.34	6.83	303.45
	Rajasthan	4.40	3.55	160.57	5.18	4.31	190.12
	Average	**5.73**	**4.33**	**196.45**	**6.45**	**4.96**	**218.61**
North East India	Arunachal Pradesh	4.73	4.12	170.12	4.73	4.12	170.12
	Assam	4.80	3.49	173.27	4.80	3.49	173.27
	Manipur	4.56	3.34	153.11	4.56	3.34	153.11
	Meghalaya	4.29	3.22	148.76	4.29	3.22	148.76
	Mizoram	8.43	6.60	285.36	8.43	6.60	285.36
	Nagaland	4.98	3.96	168.80	4.98	3.96	168.80
	Sikkim	4.52	3.78	151.07	4.52	3.78	151.07
	Tripura	5.77	4.15	195.07	5.77	4.15	195.07
	Average	**5.20**	**3.96**	**178.90**	**5.43**	**4.18**	**183.69**

areas, lack of awareness, and existing sociocultural barriers such as stigma, lack of decision-making power, societal mindsets like a breach of privacy and embarrassment (that come with exposing oneself, as well as the discomfort or pain experienced during the procedure, associated with VIA and Pap smear/cytology collection), and difficulty in accessing healthcare services [14–17]. In addition, the enormous population size of India is cited as a factor for the low cervical cancer screening [16,18].

Despite being geographically highly populous country globally (1.4 billion), China exhibits a significant difference in cervical cancer burden [1,19]. Together, China and India account for one third of all cases worldwide [1]. However, a stark disparity exists in deaths. Compared to India (79,906 deaths), China has a lower death (55,694). The predicted number of new cases in China is expected to rise by 9.5 % from 150,659 in 2022 to 160,338 in 2050, but at the same time, China may witness an overall reduction in cervical cancer mortalities in 2050. Cancer tomorrow (GLOBOCAN 2022) shows that the predicted number of new cases is expected to increase from 127,526 in 2022 to 229,056 in 2050 in India, suggesting an almost 80 % increase [1]. This trend is further mirrored by a predicted increase in cervical cancer-related deaths from 79,906 in 2022 to 153,862 by 2050, representing a 92 % increase.

The global quest to combat cervical cancer

The first nation at the forefront to eliminate cervical cancer as a public health problem is Australia. As evident from Table 3, Australia in the WPRO region has the lowest burden of cervical cancer, with incident (new case) and death ASRs of 5.3 and 1.4 per 100,000 people. In 2007, Australia integrated the HPV vaccine into their National Immunization Program (NIP) for girls aged 8–12 [20]. Subsequently, the replacement of the Pap smear (cytology) screening test with the HPV test in 2017 resulted in a significant reduction in the cumulative risk (0.5 %) of cervical cancer in the 0–75 age group [21]. Currently, 70 % of girls under age 15 in Australia are fully vaccinated with two doses of the HPV vaccine, and 80 % of women have been screened for cervical cancer using the HPV test [4].

Furthermore, in the EURO region, Switzerland takes the top spot with lower ASRs of 4.1 and 1.1 for new cases and deaths (Table 3). Currently, Switzerland's national cervical cancer management policy recommends cytological screening every three years for women aged 21–29 and HPV screening for the 30–70 age group [22]. Despite being an opportunistic screening system, nearly 80 % of women have been screened within the past five years [4,22]. Similar to Australia, 60 % of girls under 15 are fully vaccinated with two doses of the HPV vaccine [4]. The success of these policies is evident in the drastically lower cervical cancer-related mortality-to-incidence ratio (or death to new case ratio) of 0.4. Similarly, in the updated guidelines, the Centers for Disease Control and Prevention (CDC), the USA, recommend a Pap smear (cytology) test every three years for women at the age of 21–29 and primary HPV testing or co-testing at the age of 30–65 every 5 years, in addition to HPV vaccination for both girls and boys at the age of 11–12 [23]. Despite having a moderately high cervical cancer burden (new case and death ASRs of 13.6 and 7.2, Table 3), Bhutan in the Southeast Asia region presents an interesting case study in cervical cancer control policy implementation (National Cancer Control

Table 3
WHO Region-wise comparison of cervical cancer management strategies and outcomes in the top four nations with lowest incidence and mortality Age-Standardized Ratios (ASRs).

WHO Region	Country Name	Incidence		Mortality		National Policy year	Vaccination Start Year	Vaccinated Girls (Per 10)	Primary Screening Test	HPV test Adoption year	Women's Screened Ever	References
		ASRs	Total	ASRs	Total							
WPRO (21 Countries)	New Zealand	4.9	149	1.5	63	1990	2009	6	HPV	2023	97 %	[4,28]
	Australia	5.3	898	1.4	323	1991	2007	7	Pap, HPV	2017	95 %	[4,20,21]
	Vietnam	7.1	4612	3.8	2571	2016	2016	NA	HPV	2019	31 %	[4,29,30]
	Singapore	7.4	353	2.8	153	2004	2010	1	Pap, HPV	2019	67 %	[4,31]
EURO (50 Countries)	Switzerland	4.1	269	1.1	99	2003	2008	6	Pap, HPV	2021	98 %	[4,32,33]
	Malta	4.2	13	1.4	6	2011	2013	8	Pap, VIA	NA	92 %	[4,34]
	Luxembourg	4.4	22	1.7	10	1962	2008	4	Pap, HPV	NA	91 %	[4,33,35]
	Finland	4.6	179	1.3	65	1960	2013	NA	Pap, HPV	2012	90 %	[4,33,36]
PAHO (34 Countries)	USA	6.3	13,920	2.2	5932	1971	2006	6	Pap, HPV	2003	88 %	[4,37,38]
	Canada	6.6	1730	2.3	760	1960	2009	8	Pap, HPV	2003	97 %	[4,37,39]
	Costa Rica	10.6	341	4.6	167	1998	2019	7	HPV	2019	79 %	[4,40]
	Chile	11.3	1559	5.2	825	1987	2014	7	Pap	NA	79 %	[4,41,42]
EMRO (23 Countries)	Yemen	2.1	212	1.6	153	NA	NA	NA	NA	NA	10 %	[4]
	Iraq	2.2	311	1.6	216	NA	NA	NA	NA	NA	12 %	[4]
	Saudi Arabia	2.4	332	1.3	164	NA	NA	NA	NA	NA	19 %	[4]
	Syrian	2.5	206	1.3	106	NA	2011	NA	Pap	NA	5 %	[4,43]
SEARO (11 Countries)	Sri Lanka	9.2	1579	5.3	946	2015	2017	5	Pap	NA	27 %	[4,44]
	Bangladesh	11.3	9640	7	5826	2004	2023	NA	VIA	NA	7 %	[4,45,46]
	Korea	11.8	2153	6.8	1237	1999	2016	7	Pap	NA	69 %	[4,47]
	Bhutan	13.6	49	7.2	26	2000	2010	9	Pap	NA	62 %	[4,24,25,48,49]
	India	17.7	127,526	11.2	79,906	2013	NA	NA	VIA	NA	2 %	[4,9,18]
AFRO (46 Countries)	Algeria	8	1799	4.6	1013	2000	NA	NA	Pap	NA	15 %	[4,50,51]
	Niger	9.3	624	7.1	440	NA	NA	NA	NA	NA	10 %	[4]
	Mauritius	12.9	136	5.7	67	1995	2016	7	PAP, HPV	2016	42 %	[4,52–54]
	Burkina Faso	15.9	988	13	775	Yes	NA	NA	VIA	NA	6 %	[4,55]

Abbreviations: Western Pacific Regional Office (WPRO), European Regional Office (EURO), Pan American Health Organization (PAHO), Eastern Mediterranean Regional Office (EMRO), South-East Asia Regional Office (SEARO), and African Regional Office (AFRO), Not Available (NA), Papanicolaou test or Pap smear or cytology test (Pap), Human Papillomavirus test (HPV).

Program) [1,4]. Following a successful pilot HPV vaccination program in 2009, Bhutan launched its national HPV vaccination program in 2010 [24,25]. By 2020, reports indicate that Bhutan has vaccinated over 90–92 % of girls. In a progressive move, Bhutan further expanded its program to include adolescent boys in 2020 [24,25]. Recognizing the limitations of the cytology test and aligning with the WHO 90–70-90 strategy, Bhutan introduced HPV-DNA testing in 2021 [26]. This program also offers self-sampling options through the health flagship program in which a small brush is used to collect vaginal cells and fluid for sample [26]. In the HPV self-sampling method, a woman collects her own vaginal swab to test for the presence of high-risk Human Papillomavirus (hrHPV). A self-sampling kit comes with instructions to collect a sample. A brush or swab is provided, the woman inserts the swab into her vagina, rotates it as directed, and places it in a collection tube. Then the collected sample is sent to lab for testing and reporting. WHO's Global Strategy to Accelerate the Elimination of Cervical Cancer,

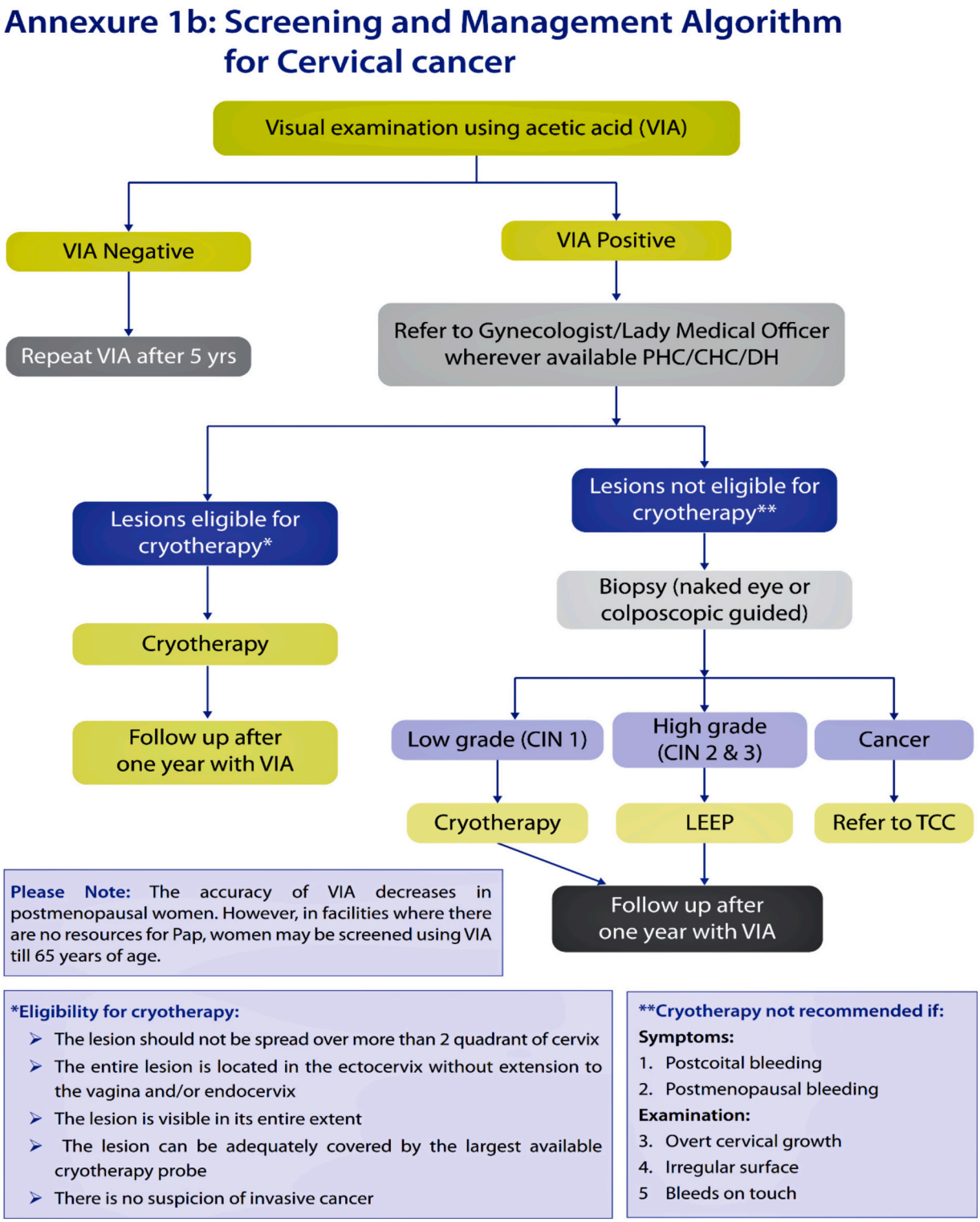

Fig. 2. India's current Screening and Management Algorithm for Cervical cancer (Source: Source: Operational Framework for Management of Common Cancers).

launched in November 2020, aims to eliminate cervical cancer as a public health problem within the century by achieving 90–70-90 targets by 2030: vaccinating 90 % of females against HPV before age 15, screening 70 % of women aged 35–45 with high-performance tests (e.g., HPV-DNA test), and ensuring 90 % of diagnosed women receive immediate and effective treatment [27].

Current cervical cancer screening practices in India

In 2016, recognizing the importance of screening and early detection in reducing the cervical cancer burden, Ministry of Health and Family Welfare (MoHFW), India, issued Operational Framework Management of Common Cancers and included cervical cancer screening with VIA testing for women aged 30–65 once every five years in the NPNCDS program (currently NP-NCD). Over the last three decades, cervical cancer screening methodologies have undergone significant transformation, from the development of the Pap smear (cytology) test in 1940 to the establishment of the link between HPV and cervical cancer in 1990. Multiple screening methods are used across India in different settings (public and private healthcare), including VIA, cytology, colposcopy, HPV-DNA testing (RT-PCR-based kits), Hybrid Capture 2 (HC2) test, etc. [56]. As per the current cervical cancer management guidelines by MoHFW, VIA remains the primary screening method in 2025, with an overall screening coverage of just 2 % (2.2 % in urban areas and 1.7 % in rural areas) as per NFHS-5 [9,13]. Unlike other methods, VIA does not require laboratory infrastructure, and it can be performed using affordable and readily available material by various health personnel (general practitioners, nurses, paramedical staff, etc.) after a short period of training [57].

A single VIA test costs approximately Indian rupee (INR) 344 in India [58]. Perhaps most crucially, the results of VIA tests are immediately (1–2 min) available, which facilitates potential same-day management for women with positive tests. Studies have shown that VIA has a varied sensitivity and lower specificity in detecting precancerous lesions due to its high subjectivity, as its results are highly dependent on interpretation skills and the examiner's level of training and expertise [6,7,57,59,60]. Furthermore, quality assurance is challenging because images of the cervix are not routinely documented in conventional VIA. Once a woman tests VIA-positive under India's national program, she is referred to secondary care hospitals for confirmation, where she undergoes colposcopy examination, biopsy if required and assessed for eligibility for ablative treatment (typically cryotherapy or thermoablation) on the same day (Fig. 2) [61]. If eligible and willing, treatment is offered immediately on-site. If women is not eligible for ablative treatment, excisional treatment is planned. The malignancies are referred to tertiary care centers. All VIA positive women are advised to visit the health facility after one year for a follow-up [9]. In practice, access to colposcopy remains very limited at and secondary care levels, contributing to delays in diagnosis and treatment [62]. Moreover, a substantial proportion of VIA-positive women do not complete the referral pathway due to logistic barriers, referral drop-outs, and infrastructure gaps; for example, in Tamil Nadu's program, approximately 41 % of VIA-positive women failed to attend colposcopy [63–65]. In sites where diagnostic follow-up is not available on the same day, loss to follow up is higher as compared to sites where colposcopy is offered immediately [58].

From awareness to action India's multi-pronged approach to combat cervical cancer

Cervical cancer is preventable; its burden can be effectively reduced by implementing three strategies. First, awareness programs are crucial and as part of the National Health Mission (NHM), India launched the NPCDCS in 2010 (currently named NP-NCD) [9]. Currently, the NCD program operates 677 district-level clinics, 266 District Day Care Centers, and 5392 community health center clinics [66]. The district-level program utilizes Accredited Social Health Activists (ASHAs) for information, education, and communication (IEC) [66]. To ensure program effectiveness, health workers undergo periodic training in screening procedures. Additionally, free diagnostic and medication facilities are being provided to patients attending these NCD clinics.

The second is vaccination against HPV infections. The vast population size may be a challenge in providing a multiple-dose HPV schedule in India. The new evidence from WHO on HPV single dose could be a game changer in the implementation and coverage of HPV vaccination in India [67]. India licensed the first bivalent vaccine (Cervarix) and quadrivalent vaccine (Gardasil)) in 2008 [18]. In 2016, Delhi became the first state in India to implement an HPV vaccination program (opportunistic) for females of the 11–13 age group [68]. Similarly, in 2016, the Punjab Government also implemented HPV vaccination in two districts with high cervical cancer burden, Bathinda and Mansa [69]. In November 2017, upon completion of this program, the first dose was received by almost 94 % of girls 11–13 age group, and the second dose was received by almost 99 % of girls [69,70]. Additionally, Sikkim in the Northeast emerged as the first state in India to achieve statewide HPV vaccination for girls aged 9–13 years in 2018 [70,71]. By the end of that year, nearly 97 % of girls in this age group had received both doses of the vaccine [71]. Based on these studies and promising results from several other pilot studies conducted in various Indian states in February 2024, the Union Finance Minister of India announced a proposal to introduce nationwide HPV vaccination for girls aged 9–14 to prevent cervical cancer in a phased manner [72].

Third is the implementation of a highly sensitive primary screening test (HPV-DNA test) [73]. Numerous studies demonstrate that HPV-DNA testing significantly reduces advanced cervical cancers and deaths [8,58]. In a 2021 report by the Health Technology Assessment India (HTA1n) MoHFW, India, assessing the effectiveness of VIA and HPV-DNA screening across screening intervals of 3, 5, and 10 years, it was found that both VIA and HPV-DNA testing can lower cervical cancer incidence and deaths [58]. HPV-DNA screening consistently outperforms other methods, achieving the highest percentage decrease in both cervical cancer new cases and deaths across all frequencies [58]. HPV-DNA screening conducted at intervals of 5 years leads to a decrease in new cases by 42.5 % and deaths by 56 % compared to VIA tests, which show a decrease of 36 % in new cases and 51 % in deaths [58]. However, in India, a concern about significant price disparity exists between VIA and HPV-DNA testing [7,58]. The quoted cost of INR 1500–3000 for the HPV-DNA test in different studies and news articles is for the kits available in the market for individual testing [74]. The significant cost of the HPV-DNA test available in the market, relative to India's per capita net national income (₹98,374 in 2022–23), seems to make HPV testing out of reach for many people [75]. On the contrary, as per the HTA1n, India report, the calculated cost for VIA is INR 344, and HPV is INR 980 in India [58]. However, the cost of HPV-DNA testing could be further reduced with increased demand for screening, especially if implemented on a large scale at a programmatic level. In our past study to evaluate the feasibility and acceptability of HPV-DNA testing in the government program, the state government of Sikkim, India, negotiated and procured the testing kits and reagents at a much more affordable, subsidized cost from the manufacturers [76]. Efforts are being made by the Indian government to develop a low-cost, indigenous, portable—colposcopy, thermal ablators, and HPV-DNA detection kit to strengthen the cervical cancer screening program in resource-limited settings in India [17,77,78].

Discussion

In this study, we found that the burden of cervical cancer varies significantly across WHO regions, with the AFRO and SEARO regions shouldering the highest incidents (new cases) and deaths. India is the largest contributor to the cervical cancer burden in SEARO and ranks third globally in terms of new cases and second in deaths. The analysis of GBD data for India from 2012 to 2019 reveals a significant increase in

cervical cancer deaths, with notable regional variations. The South Region bears the highest burden, followed by the West Region. While overall mortalities increased from 2012 to 2019, the annual percentage change slowed between 2016 and 2019. Alarmingly, the national average for deaths, YLDs, and DALYs have increased in 2016, emphasizing the need to improve cervical cancer screening and management services. Cervical cancer screening has advanced significantly over the past decades. The introduction of the Pap smear in the 1940s revolutionized screening, followed by liquid-based cytology and HPV testing in later years. Developed nations have lower cervical cancer prevalence than developing countries within the same WHO regions, largely because of widespread, high-quality screening services (regular HPV testing or HPV testing combined with repeated cytology), accessible self-sampling options, timely referrals, and well-resourced health systems that ensure rigorous follow-up and treatment completion [4,20,21,79,80].

One reason developing countries have been successful in reducing the burden of cervical cancer is the integration of policies similar to the 90–70-90 strategy into their cervical cancer control policies [4]. Despite being the world's fifth-largest economy, India ranks fourth in cervical cancer new cases and death ASRs and contributes the highest number of cervical cancer-related deaths [1,81]. The GLOBOCAN 2022 data suggests that in the upcoming 25–28 years, India may face a formidable cervical cancer challenge and require fast action to achieve the WHO Global aims to eliminate cervical cancer [1,2]. Upon HPV exposure, nearly 70 % of new infections clear within a year, and 91 % within two years [82]. Only 25 % of high-risk HPV infections progress to cervical intraepithelial neoplasia, and about 1 % develop into invasive cervical cancer [82]. Given India's population of 1.4 billion, with approximately 472.5 million females in the 15–64 age group, approximately 67.5 % of this population falls within the 15–64 age group [83,84]. Considering India's overall male-to-female ratio of 1.07, this translates to roughly 472.5 million females being in the 15–64 age group, and 60–80 % of these women may encounter HPV infection in the coming years [85]. On vaccination, girls are protected from high-risk HPV infection and hence protected from acquiring cervical cancer [86]. Recognizing this, in February 2024, India announced the rollout of a nationwide program for cervical cancer vaccination using India's first indigenous HPV vaccine, Cervavac, for girls aged 9 to 14, which is a landmark initiative to reduce the cervical cancer burden in India [72]. Nevertheless, as of November 2025, the HPV vaccination is yet to be implemented. However, if all eligible girls are vaccinated in 2025–2028 at approximately ages 10–14 with 90 % coverage and receive lifelong protection from a single or two-dose HPV vaccination, the prevalence of HPV16 and HPV18 infections could be reduced by 97 % over the next 50 years. This reduction will significantly lower the risk of cervical cancer, particularly among women aged 30–49, who typically enter the high-risk age group about 15–20 years after vaccination (around 2040 to 2045). As a result, the impact on cervical cancer new cases may be observed as early as 2040, with a more noticeable reduction in burden expected by 2045. Additionally, the lifetime risk of cervical cancer could be reduced by 71–78 % [87]. In addition to its major impact on cervical cancer, HPV vaccination may prevent the HPV-attributable fraction of oropharyngeal cancers (type of head and neck cancer); about 63.2 % of male (if male children are also vaccinated) and 3.6 % of female cases, offering a modest but valuable public-health benefit [88,89].

Despite the established causal link between HPV infection and cervical cancer, India's national guidelines in 2016 implemented Visual Inspection with Acetic Acid (VIA) for primary screening. This strategy was a pragmatic response to the country's resource limitations and infrastructural challenges, serving as a transitional solution until the availability of cost-effective and dependable HPV diagnostic tests [90]. Despite the completion of a decade of implementing VIA as a primary screening test, the cervical cancer burden has not changed much in India. The NHFS5 survey indicates that overall screening coverage in India is just 0.3–2 %, with high death rates in many states. Inadequate screening coverage and lack of follow-up after VIA testing are major contributing factors, particularly in rural and remote areas, due to limited infrastructure, low awareness, and sociocultural barriers like privacy concerns and embarrassment [13,17,63–65,90]. This gap is at most concerning, as many of the VIA screen-positive cases fail to undergo colposcopy for confirmation or receive timely treatment interventions of utmost importance to lower the increased prevalence of high-grade lesions or cervical cancer. Tamil Nadu state, which introduced VIA screening in 2011, reported the highest screening coverage (9.8 %) in the NHFS5 report; yet its cervical cancer deaths increased from 12.5 per 100,000 women in 2012 to 12.5 in 2019. This shows that the high subjectivity, varied sensitivity, and specificity of VIA in detecting precancerous lesions are also major reasons for the increased cervical cancer burden in India [6,13,65]. Because VIA only examines the ectocervix region, it increases the risk of missing glandular intraepithelial lesions or squamous cell lesions within the cervical canal, leading to missed detection of endocervical lesions [59]. Additionally, the effectiveness of VIA also decreases in older women [59].

The successful implementation of any program hinges on the quality of services offered and the improved participation of the beneficiaries. The specific hurdles in sampling and women's willingness to get screened could be addressed by, allowing women to self-collect samples for HPV testing, which can significantly enhance participation. The subjectivity, sensitivity, and specificity issues can be addressed by adopting HPV-DNA testing, which is a high-performance test. HPV-DNA testing detects infections more effectively than other methods, reducing cervical cancer cases to approximately 1 per 100,000 women annually compared to 4 per 100,000 with other tests [56,91,92]. Additionally, it is a well-established fact that self-sampling is possible in HPV-DNA testing, with ensured privacy, more women may be encouraged to undergo cervical cancer screening, whereas self-sampling strategies are not possible for both VIA and Pap smear tests [57,76,93]. However, like other tests, it carries a risk of over-diagnosis, as transient HPV infections that would have cleared naturally may be detected [73]. Awareness initiatives should go beyond medical education and address societal prejudices by involving local officials and female health workers to build trust and encourage women to undergo screening and follow-up tests.

The Indian government has significantly increased healthcare expenditure over the past five years, with per capita spending rising from INR 1108 in 2014–15 to INR 3169 in 2021–22 [94]. On a roadmap to become a developed nation, the government of India has introduced a range of health care schemes such as—Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB-PMJAY), National Health Protection Scheme (NHPS), National Dialysis Programme, National Cancer Control Programme, etc. to improve access to affordable and quality healthcare for its citizens [95,96]. Strengthening the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD) by incorporating HPV-DNA testing as the primary screening method and allowing self-sampling could significantly enhance screening coverage over time. While HPV-DNA testing may seem expensive initially, WHO recommends it more cost-effective in the long run, as early detection reduces the burden of treating advanced cervical cancer [58,97]. Public-private partnerships and NGOs can help address cost barriers, raise awareness, and offer mobile screening services in underserved areas. The successful implementation of any screening program depends on accessibility, affordability, and social acceptance. Success stories from Australia, Denmark, Finland, and Bhutan demonstrate that sustained HPV-DNA screening significantly reduces cervical cancer new cases. Expanding HPV-DNA testing in low-resource settings poses logistical and financial challenges, however, India's response to the COVID-19 pandemic was indeed commendable, and the leveraged health infrastructure, established screening laboratories, can be used for HPV-DNA testing for screening and diagnosis. Leveraging existing screening labs, mobile screening units, and telemedicine consultations with urban gynecologists could enhance coverage, while ongoing research into cost-effective, indigenous HPV testing kits could further support screening

efforts [17,77,78]. Until then, available HPV screening kits should be used to bridge the gap, and innovative early detection and treatment methods must be explored to reduce India's cervical cancer burden.

Conclusion

In conclusion, the burden of cervical cancer in India is significant, with the South Region being the most affected. In recent years, India has made efforts to reduce the burden of cervical cancer. However, the current VIA-based screening program has limitations, and despite these efforts, national averages for mortality (death), YLDs, and DALYs have increased. India must prioritize widespread, highly precise testing with a self-sampling option as a primary screening method to address major limitations of the VIA test and to reduce the cervical cancer burden. HPV-DNA testing can address the underlying issues associated with the current VIA test. Furthermore, India can leverage the health infrastructure established for the COVID-19 pandemic to increase cervical cancer screening, early diagnosis, and strengthen existing government-funded health programs. Despite challenges, leveraging existing health infrastructure, public-private partnerships, and research efforts can make HPV-DNA testing more accessible and cost-effective.

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CRediT authorship contribution statement

Khushwant Singh: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ashoo Grover:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Kavitha Dhanasekaran:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

Authors do not have any competing interests.

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Appendix A. Supplementary data

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